

Λ -EG: A Screened Scalar–Tensor Framework for Low-Acceleration Gravitational Phenomenology

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Abstract

We present Λ -EG (Lambda Emergent Gravity), a covariant scalar–tensor framework in which a k-essence field ϕ recovers General Relativity in screened high-acceleration environments while reproducing modified dynamics in the low-acceleration regime relevant to galactic phenomenology. The critical acceleration scale

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12} \text{ m s}^{-2}$$

is motivated, without free parameters, primarily by the spectral geometry of the De Sitter horizon: the first non-zero Beltrami–Laplace eigenvalue on S^3 ($\lambda_1 = 3$) fixes the $\sqrt{3}$, while the $4\pi^2$ factor follows from the horizon heat-kernel measure. The Dirac–Lichnerowicz curvature term $R/4$ on S^4 yields the same value under a common curvature-radius normalization, as a related geometric consistency check. A complete variational principle unifying these ingredients remains an open aspect of the framework.

Starting from a k-essence action $\mathcal{L}_\phi = -a_*^2 F(X)$ with $X = (\partial\phi)^2/a_*^2$, we derive three mutually consistent field equations: (i) the covariant k-essence equation; (ii) the modified Einstein equations with k-essence source; (iii) the non-relativistic limit of a screened scalar–tensor structure, in which the deep-MOND relation $V^4 = GM_{\text{bar}} a_0$ (with $a_0 = 4\pi g_{\text{crit}}$) emerges asymptotically when the scalar contribution dominates, while the Newtonian regime is recovered through screening. The pure kinetic branch $F(X) \propto X^{3/2}$ corresponds to the deep-MOND (infrared) limit; the complete kinetic function, used consistently throughout, interpolates to Newtonian dynamics at high acceleration and yields $\mu(x) = x/\sqrt{1+x^2}$. In dense environments a density-dependent screening potential traps the field near a local minimum, suppressing the scalar force through a thin-shell mechanism that satisfies Cassini PPN bounds ($|\gamma-1| \lesssim 2.3 \times 10^{-5}$) for sufficiently large coupling ($\alpha \gtrsim 8 \times 10^6$, with $F_\phi/F_N \propto \alpha^{-5/3}$). For the asymptotic kinetic branch $F(X) \propto X^{3/2}$, scalar perturbations propagate at constant subluminal speed $c_s^2 = 1/2$, ensuring strict hyperbolicity. A topological transition radius $r_t = \sqrt{GM/g_{\text{crit}}}$ separates the screened and unscreened dynamical domains.

Observational validation against 184 galaxies from the SPARC and LITTLE THINGS catalogs is reported in the companion papers [2,3]. The present paper provides the complete field-theoretic foundation. Open problems are stated explicitly.

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1 Introduction

General Relativity (GR) is confirmed to high precision from Solar-System to compact-object scales. At galactic and larger scales, the standard Λ CDM paradigm invokes non-baryonic dark matter to account for observed mass discrepancies. Despite its considerable empirical success, the microphysical identity of dark matter remains unknown.

Several observational regularities motivate continued investigation of the gravitational sector. The radial acceleration relation [11], the baryonic Tully–Fisher relation (BTFR) [10], and the flatness of rotation curves in low-surface-brightness (LSB) galaxies [12] all suggest that galactic dynamics contains structure beyond purely Newtonian expectations when baryonic mass alone is considered. The small intrinsic scatter ($\sigma \approx 0.07$ dex) of the BTFR is particularly constraining: it implies a tight coupling between baryonic mass and asymptotic orbital velocity that is difficult to accommodate naturally within stochastic dark-matter halo formation [10].

Modified-gravity proposals including MOND [4], AQUAL [5], TeVeS, screened scalar-tensor theories [7,8], and emergent gravity scenarios [9] have addressed different aspects of this problem. The present paper introduces Λ -EG (Lambda Emergent Gravity), a compact covariant framework combining three properties:

- (i) recovery of GR in screened high-acceleration environments;
- (ii) modified dynamics in the low-acceleration regime, reproducing flat rotation curves and BTFR scaling;
- (iii) a critical acceleration scale g_{crit} derived from the spectral geometry of the De Sitter horizon, linking infrared gravitational phenomenology to the cosmological background without free parameters.

Figure 1 summarizes the logical structure of the framework, from the action to the observational phenomenology. The fundamental dynamical object of Λ -EG is the scalar field ϕ : the spectral construction of Section 3 fixes the critical scale, while decoherence in the De Sitter background provides its physical interpretation. What distinguishes Λ -EG from a scalar–tensor theory with a phenomenologically chosen potential is precisely this: the low-acceleration scale g_{crit} is *not* a free parameter tuned to galaxies, but is fixed by the spectral geometry of the cosmological horizon together with the present Hubble rate H_0 . The empirical MOND scale thereby acquires a geometric origin, $a_0 = 4\pi g_{\text{crit}}$.

This geometric origin yields a prediction that distinguishes Λ -EG from MOND. In MOND the acceleration scale a_0 is a universal constant; in Λ -EG it inherits the cosmological background, $g_{\text{crit}}(z) \propto H(z)$, so the zero-point of the BTFR evolves with redshift as $V \propto [H(z)/H_0]^{1/4}$. This evolution is *absent* in standard MOND and is testable with high-redshift kinematics (e.g. JWST/JADES); a second, independent discriminator is the acceleration scale probed by wide binary stars, which separates g_{crit} from a_0 (Gaia). These provide concrete tests where Λ -EG and MOND diverge, rather than merely reproducing shared low-redshift phenomenology. We flag at the outset that $g_{\text{crit}}(z) \propto H(z)$ is the *instantaneous–de Sitter* reading of the spectral derivation of Section 3: there g_{crit} is set by the curvature of the horizon, and the identity $R = 12H^2$ that turns curvature into Hubble rate is exact in de Sitter space only. The covariant alternative is discussed where the evolution is derived; it is the reading favoured by the data, but it is a reading and not a consequence.

The field-theoretic structure and geometric derivation of g_{crit} are the primary content of this paper. Observational validation against galaxy rotation curves and predictions for the cosmological evolution of the BTFR are presented in the companion papers [2,3], to which we refer for all numerical confrontation with data.

We adopt units $c = 1$ where convenient and restore factors of c in final expressions. The signature is $(-, +, +, +)$. We use $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck 2020 [21]) throughout.

2 The Λ -EG Action and Field Equations

2.1 Action

We consider a real scalar field ϕ minimally coupled to gravity, with a density-dependent screening potential $V(\phi, \rho)$:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{\Lambda}{8\pi G} - a_*^2 F(X) - V(\phi, \rho) \right] + S_{\text{mat}}, \quad (1)$$

where

$$X = \frac{(\partial\phi)^2}{a_*^2} = \frac{g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi}{a_*^2} \quad (2)$$

is the dimensionless kinetic invariant, and $a_* \equiv g_{\text{crit}}$ is the critical acceleration scale fixed by De Sitter spectral geometry (Section 3). The normalization by a_*^2 is not optional: without it the vacuum energy density develops as $\rho_\phi \sim H_0^3/G$ instead of the physical $\rho_\Lambda \sim H_0^2/G$, yielding an inconsistent cosmological limit.

The action (1) belongs to the k-essence class [13], with a_* fixed by spectral geometry rather than fitted to data. The screening potential $V(\phi, \rho)$ is specified in Section 4.

2.2 Field equations

Variation of (1) with respect to $g^{\mu\nu}$ gives the modified Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(\text{mat})} + T_{\mu\nu}^{(\phi)} \right), \quad (3)$$

with scalar stress-energy tensor

$$T_{\mu\nu}^{(\phi)} = 2F'(X) \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} a_*^2 F(X) - g_{\mu\nu} V, \quad (4)$$

where $F'(X) \equiv dF/dX$. Variation with respect to ϕ yields the covariant k-essence equation:

$$\nabla_\mu [F'(X) \partial^\mu \phi] = \frac{\partial V}{\partial \phi}. \quad (5)$$

In the non-relativistic quasi-static limit (Section 2.5), with $x \equiv |\nabla\Phi|/a_*$ and Φ the Newtonian potential, the three field equations reduce to a mutually consistent set:

$$(\text{k-essence NR}): \quad \nabla \cdot [F'(x^2) \nabla \Phi] = 4\pi G \rho, \quad (6)$$

$$(\text{modified Poisson}): \quad \nabla \cdot [\mu(|\nabla\Phi|/a_*) \nabla \Phi] = 4\pi G \rho, \quad \mu(x) = F'(x^2), \quad (7)$$

$$(\text{BTFR limit}): \quad V^4 = GM_{\text{bar}} a_0, \quad a_0 = 4\pi g_{\text{crit}}, \quad (8)$$

where equation (7) is the AQUAL equation of Bekenstein and Milgrom [5] and equation (8) holds in the deep collective regime $x \ll 1$.

2.3 Fixing the kinetic function

The MOND interpolating condition $\mu(x) \sim x$ for $x \ll 1$ fixes the deep-MOND (infrared) branch of the kinetic function,

$$F'(X) \sim X^{1/2} \implies F_{\text{IR}}(X) = \frac{2}{3} X^{3/2}, \quad \mu(x) = F'_{\text{IR}}(x^2) = x, \quad (9)$$

which reproduces asymptotically flat rotation curves at low acceleration. This pure cubic form captures *only* the deep-MOND limit and gives $\mu(x) = x$; it does not by itself recover Newtonian dynamics. The complete kinetic function used throughout,

$$F(X) = \sqrt{X(1+X)} - \text{arcsinh}\sqrt{X}, \quad \mu(x) = F'(x^2) = \frac{x}{\sqrt{1+x^2}}, \quad (10)$$

interpolates smoothly between this limit and Newtonian dynamics, with $F(X) \rightarrow X$ and $\mu \rightarrow 1$ at $x \gg 1$ (Figure 2). The dynamical selection of this branch is discussed in the following subsection.

2.4 Dynamical selection of the kinetic branch

The functional form $F(X) \propto X^{3/2}$ identified in equation (9) is not an arbitrary choice imposed to reproduce MOND phenomenology. It emerges as the stable branch dynamically selected by the evolution of the scalar field in the low-acceleration regime.

To see this, consider the vacuum ($\rho \rightarrow 0$), quasi-static, spherically symmetric limit of the k-essence equation (5):

$$\frac{1}{r^2} \frac{d}{dr} (r^2 F_X \phi') = 0, \quad (11)$$

which integrates to

$$F_X \phi' = \frac{C}{r^2}, \quad (12)$$

where $F_X \equiv dF/dX$ and C is an integration constant fixed by the baryonic mass distribution. Defining $X = (\phi')^2$, this relation becomes

$$F_X \sqrt{X} \propto \frac{1}{r^2}. \quad (13)$$

For a general power-law kinetic function $F(X) \sim X^n$, one has $F_X \sim X^{n-1}$, leading to

$$X^{n-1/2} \propto \frac{1}{r^2}. \quad (14)$$

Solving for the radial dependence,

$$X(r) \propto r^{-\frac{2}{n-1/2}}, \quad \phi'(r) \propto r^{-\frac{1}{n-1/2}}. \quad (15)$$

Asymptotically flat rotation curves require $\phi'(r) \propto r^{-1}$, which uniquely fixes

$$n = \frac{3}{2}. \quad (16)$$

This establishes that $F(X) \propto X^{3/2}$ is the only kinetic branch compatible with stationary solutions exhibiting flat circular velocities.

To address whether this branch is dynamically selected, consider the non-equilibrium scalar-field dynamics derived in Paper II [3], which introduces a dissipative term associated with cosmological expansion. In the comoving frame, perturbations δX around a background solution satisfy a schematic evolution equation of the form

$$\delta X'' + \Gamma \delta X' + \mathcal{M}^2(X) \delta X = 0, \quad (17)$$

where $\Gamma \sim 3H_0 > 0$ represents Hubble friction and $\mathcal{M}^2$ is an effective stability operator determined by $F(X)$. The presence of positive friction implies that unstable or non-stationary branches decay toward dynamically stable configurations. Evaluating the effective restoring term $\mathcal{M}^2$ for perturbations around power-law solutions shows that only the branch $n = 3/2$ yields a non-negative operator, ensuring that $\delta X \rightarrow 0$ at large radii. Deviations from this branch either decay toward it or fail to produce stationary solutions compatible with finite-energy boundary conditions.

Therefore, the asymptotic form

$$F(X) = \frac{2}{3} X^{3/2} \quad (18)$$

is dynamically selected as an attractor of the scalar-field evolution in the infrared regime. The combined requirements of (i) regularity, (ii) finite energy, (iii) flat rotation curves, and (iv) dissipative relaxation uniquely select the $X^{3/2}$ branch, removing the apparent degeneracy in the choice of the kinetic function.

Remark 2.1. *A complete analytic derivation of the stability operator $\mathcal{M}^2(X)$ confirming that $n = 3/2$ is the unique non-negative branch is deferred to future work. The selection is confirmed observationally by the dual falsification test of Paper I [2], which rules out $n \neq 3/2$ at median velocity error $< 20\%$ across the collective galaxy subsample.*

2.5 Non-relativistic limit and screened scalar–tensor structure

We now derive the non-relativistic (NR) limit of the covariant field equations explicitly, showing how the deep-MOND (AQUAL) behaviour emerges asymptotically within a screened scalar–tensor structure.

We consider weak-field, quasi-static configurations around Minkowski space,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1, \quad (19)$$

with the standard identification of the Newtonian potential,

$$h_{00} = -2\Phi_N, \quad h_{ij} = 2\Phi_N\delta_{ij}. \quad (20)$$

In this regime, time derivatives are subdominant compared to spatial gradients, $\partial_t\phi \ll \nabla\phi$, and the kinetic scalar reduces to

$$X = \frac{(\nabla\phi)^2}{a_*^2}. \quad (21)$$

The covariant k-essence equation (5) then reduces to

$$\nabla \cdot (F_X \nabla\phi) = \frac{\partial V_{\text{eff}}}{\partial\phi}. \quad (22)$$

In the low-density regime outside screened regions, the effective potential gradient is negligible,

$$\frac{\partial V_{\text{eff}}}{\partial\phi} \approx 0, \quad (23)$$

leading to

$$\nabla \cdot (F_X \nabla\phi) = 0. \quad (24)$$

The physical gravitational potential is defined as the sum of the Newtonian contribution and the scalar-field correction,

$$\Phi = \Phi_N + c^2\alpha\phi, \quad (25)$$

so that test-particle acceleration is governed by $\mathbf{g} = -\nabla\Phi$. Using the baryonic Poisson equation, $\nabla^2\Phi_N = 4\pi G\rho$, and combining both contributions, we obtain

$$\nabla \cdot [\nabla\Phi_N + c^2\alpha F_X \nabla\phi] = 4\pi G\rho. \quad (26)$$

The full non-relativistic equation obtained above is *not* identical to the AQUAL equation: it retains separate Newtonian and scalar contributions. In the low-acceleration (deep-MOND) regime, however, the scalar contribution dominates, $|\nabla\Phi| \rightarrow |\nabla(c^2\alpha\phi)|$, and defining the interpolating function

$$\mu\left(\frac{|\nabla\Phi|}{a_*}\right) \equiv F_X, \quad (27)$$

the equation reduces asymptotically to the AQUAL form

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi|}{a_*}\right) \nabla\Phi \right] = 4\pi G\rho, \quad (28)$$

of Bekenstein and Milgrom [5]. The standard AQUAL behaviour thus emerges as the deep-MOND asymptotic limit of a screened scalar–tensor structure, not as an exact identity of the full non-relativistic equations; the Newtonian regime is recovered through the screening mechanism, which suppresses the scalar contribution at high acceleration. In this sense the MOND-like modification arises dynamically from the covariant theory rather than being imposed phenomenologically.

3 Origin of the Critical Scale: Spectral Geometry of De Sitter

The value of a_* is not a free parameter. The relevant geometry is not chosen among manifolds: S^3 is the spatial section of De Sitter space, the maximally symmetric late-time attractor of an accelerating universe, so the construction is fixed by the physical vacuum rather than by convenience. We obtain the scale *primarily* from the scalar spectral gap on this S^3 (Section 3.1), with an *independent* geometric consistency check from the curvature term on S^4 (Section 3.2); both give

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12} \text{ m s}^{-2} \quad (29)$$

with $H_0 = 67.4 \text{ km s}^{-1}\text{Mpc}^{-1}$ [21]. The relevant infrared scale is the *present* Hubble rate because the coherence length set by the De Sitter thermal bath equals the current horizon radius, $\ell_{\text{dec}} \simeq c/H_0$; consequently $g_{\text{crit}}(z) \propto H(z)$, the origin of the redshift dependence tested at high z . Why H_0 takes its observed value is the cosmological-constant problem and is not addressed here.

3.1 Primary derivation: scalar spectral gap on S^3

The De Sitter horizon has spatial section S^3 with Hubble radius $R_H = c/H_0$. The Beltrami–Laplace operator Δ_{S^3} has eigenvalues

$$\lambda_\ell^{S^3} = \ell(\ell + 2), \quad \ell = 0, 1, 2, \dots \quad (30)$$

The $\ell = 0$ mode is the constant function: it carries zero energy and provides no dynamical scale. The first non-trivial mode is $\ell = 1$ by topology, with

$$\lambda_1^{S^3} = 3 \implies \sqrt{\lambda_1^{S^3}} = \sqrt{3}. \quad (31)$$

The factor $4\pi^2 = 2 \text{Vol}(S^3)$ arises from the horizon heat-kernel measure ($\text{Vol}(S^3) = 2\pi^2$), reflecting the four-dimensionality of De Sitter spacetime. Combining the spectral gap $\sqrt{\lambda_1^{S^3}} = \sqrt{3}$ with this measure fixes the scale,

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}}. \quad (32)$$

This spectral/heat-kernel construction is presented as the primary route *motivating* the numerical factor; a complete variational principle unifying the spectral gap and the measure into a single derivation remains an open aspect of the framework. Three independent arguments establish the inevitability of $\ell = 1$: (a) topological (it is the first mode with nonzero eigenvalue); (b) Bunch–Davies projection from S^4 onto S^3 (the $\ell_4 = 0$ mode does not project onto $\ell_3 = 1$, while $\ell_4 = 1$ does, verified numerically); (c) geodesic embedding ($S^3 \subset S^4$ is totally geodesic, so the induced operator carries eigenvalues $\lambda_\ell^{S^3}$ without extrinsic correction). The resulting number is therefore rigid: a different manifold or a different mode would change it (the first non-zero Laplace eigenvalues on S^2 and S^4 are 2 and 4, not 3), so $\sqrt{3}$ is specific to the $\ell = 1$ mode on the S^3 section of De Sitter and is not adjustable. Being a topological spectral gap, it is robust: it is insensitive to the detailed form of the screening potential or of the coupling, which affect the phenomenology but not the value of g_{crit} .

3.2 Independent consistency check: Dirac curvature term on S^4

The Lichnerowicz–Weitzenböck identity for the *Dirac* operator (acting on spinor sections) on a Riemannian manifold is

$$\mathcal{D}^2 = -\square + \frac{R}{4}, \quad (33)$$

where R is the Ricci scalar. On the unit S^4 , $R = 12$, so the curvature contribution is $R/4 = 3$. Under the common curvature-radius normalization (S^3 the equatorial section of S^4 , both of Hubble radius R_H), this curvature term coincides numerically with the scalar spectral gap $\lambda_1^{S^3} = 3$ of Route 1, giving $m_{\text{eff}}^2 = 3$ and $m_{\text{eff}} = \sqrt{3}$. We stress that the $R/4$ term is the curvature contribution proper to the Dirac operator and does *not* apply to a minimally coupled scalar field; Route 2 is therefore offered as a related geometric quantity providing an internal consistency check on the value obtained from the scalar spectral gap of Route 1, which we take as the primary origin of the $\sqrt{3}$. In physical units, the curvature-induced mass is

$$\frac{m_{\text{eff}} c^2}{\hbar} = \frac{\sqrt{3} H_0}{c} \implies a_* = \frac{m_{\text{eff}} c^2}{\hbar} \cdot c = \frac{\sqrt{3} H_0 c}{c} = \frac{c H_0}{\sqrt{3}}, \quad (34)$$

and upon inclusion of the $4\pi^2$ volume factor from Route 1:

$$g_{\text{crit}} = \frac{c H_0}{4\pi^2 \sqrt{3}}. \quad (35)$$

The numerical identity $\lambda_1^{S^3} = R_{S^4}/4 = 3$ is not a coincidence: S^3 is the spatial section of S^4 , and the curvature felt by the field through the Dirac operator is encoded in the same eigenvalue governing the horizon's smallest oscillation.

3.3 Quantum bridge: Bunch-Davies vacuum

The Gibbons–Hawking temperature of the De Sitter horizon is $T_{\text{GH}} = \hbar H_0 / (2\pi k_B)$ [14]. The corresponding horizon action quantum is

$$S_{\text{GH}} = \frac{k_B T_{\text{GH}}}{H_0} = \frac{\hbar}{2\pi}. \quad (36)$$

The Wightman propagator of the Bunch-Davies vacuum [15] yields, in the infrared limit, a dissipation coefficient

$$\gamma_{\text{BD}} = \frac{H_0}{4\pi^2}, \quad (37)$$

physically distinct from the classical Hubble friction $3H_0$ derived from the FLRW Lagrangian in Paper II [3]. The two coexist in the full framework and differ by a factor $12\pi^2 \approx 118$.

Combining γ_{BD} with the geometric factor $c/\sqrt{3}$ from Route 1:

$$g_{\text{crit}} = \gamma_{\text{BD}} \cdot \frac{c}{\sqrt{3}} = \frac{H_0}{4\pi^2} \cdot \frac{c}{\sqrt{3}} = \frac{c H_0}{4\pi^2 \sqrt{3}}. \quad (38)$$

The agreement between the geometric derivation (Section 3.1), the Dirac–Lichnerowicz route (Section 3.2), and this quantum bridge is an internal consistency check: all three employ independent mathematical structures yet converge to the same expression.

Remark 3.1 (Quantum origin of a classical expression). *Although the final result $g_{\text{crit}} = c H_0 / (4\pi^2 \sqrt{3})$ contains only the classical constants c and H_0 , its derivation via the Bunch-Davies vacuum is inherently quantum: $\hbar$ enters through the Gibbons-Hawking temperature $T_{\text{GH}} = \hbar H_0 / (2\pi k_B)$ and cancels in the decoherence rate $k_B T_{\text{GH}} / \hbar = H_0 / (2\pi)$. This cancellation mirrors the structure of Hawking radiation, where the quantum temperature $T_H \propto \hbar$ combined with the transition rate $\propto T_H / \hbar$ yields a classical-looking thermal rate. Equation (29) is therefore quantum-classical in character: quantum in origin, classical in expression.*

4 Screening Sector and PPN Constraints

Theorem 4.1 (Separation Theorem). *The critical acceleration factorises as $g_{\text{crit}} = S \times K$, where*

- $S = cH_0$ is the scale factor: the unique acceleration constructible from the horizon constants c and H_0 . It evolves cosmologically as $S(z) = cH(z)$.
- $K = 1/(4\pi^2\sqrt{3})$ is the kernel factor: a pure dimensionless number determined entirely by the geometry of the Bunch-Davies propagator G^+ in four-dimensional De Sitter spacetime. It is universal and independent of all orbital parameters and redshift.

This factorisation separates cosmological evolution (encoded in S) from the geometric structure of the vacuum (encoded in K), explaining why $g_{\text{crit}}(z) \propto H(z)$ while the spectral derivation of Section 3 is z -independent. It also clarifies the quantum-classical duality noted in Section 3.3: $\hbar$ cancels structurally in K through the ratio $k_B T_{\text{GH}}/\hbar = H_0/(2\pi)$.

4.1 Density-dependent effective potential

The screening potential in action (1) takes the form

$$V_{\text{eff}}(\phi, \rho) = \Lambda_0^4 \left[\frac{1}{2}(\phi - 1)^2 + \frac{\xi}{\phi^n} \right] + \alpha \phi \rho c^2, \quad (39)$$

where Λ_0 is the structural energy scale, α the baryonic coupling, and n, ξ dimensionless parameters. Dimensional consistency requires

$$\Lambda_0^4 \xi \approx \rho_\Lambda c^2, \quad (40)$$

where $\rho_\Lambda \approx 10^{-29} \text{ g cm}^{-3}$ is the cosmological vacuum energy density. This condition ties the screening scale to the observed dark energy density without additional fine-tuning.

In dense environments the effective potential acquires a density-dependent minimum:

$$\left. \frac{\partial V_{\text{eff}}}{\partial \phi} \right|_{\phi_{\text{min}}} = 0 \implies \phi_{\text{min}}(\rho) \sim \left(\frac{\rho_\Lambda}{\alpha \rho} \right)^{1/(n+1)}. \quad (41)$$

For $n = 2$, $\phi_{\text{min}} \sim (\rho_\Lambda/\alpha\rho)^{1/3}$. At galactic halo densities $\rho \sim 10^{-24} \text{ g cm}^{-3}$, the field takes a value $\phi_{\text{min}} \sim 10^{-2}$ – 10^{-4} (depending on α), while in the cosmological void ($\rho \sim 10^{-30} \text{ g cm}^{-3}$) it approaches unity. Figure 3 shows this profile across the relevant density range.

4.2 Thin-shell mechanism

Inside a compact object the field is trapped at $\phi_{\text{min}}(\rho_{\text{int}})$, and only a thin shell of thickness $\Delta R \ll R$ near the surface contributes appreciably to the exterior scalar force. The force ratio between the scalar and Newtonian components is [7]

$$\frac{F_\phi}{F_N} \simeq \frac{\Delta\phi}{2\Phi_N}, \quad (42)$$

where Φ_N is the Newtonian surface potential and $\Delta\phi \equiv \phi_{\text{out}} - \phi_{\text{min}}$. Figure 4 illustrates the radial profile $\phi(r)$ for a solar-type object, showing the strongly screened interior, the thin-shell transition, and the exterior relaxation.

4.3 PPN constraint from Cassini

Solar-System precision tests require $|\gamma - 1| \lesssim 2.3 \times 10^{-5}$, where γ is the PPN parameter [16, 17]. This translates into

$$\frac{F_\phi}{F_N} = \frac{\Delta\phi}{2\Phi_N} \ll 2.3 \times 10^{-5}. \quad (43)$$

With Newtonian surface potential $\Phi_{N,\odot} \approx 2 \times 10^{-6}$, the Cassini bound requires $\Delta\phi \lesssim 10^{-10} - 10^{-11}$.

To estimate $\Delta\phi$: at the solar interior density $\rho_{\text{int}} \sim 10^{-1} \text{ g cm}^{-3}$ and with $n = 2$, the field minimum satisfies $\phi_{\text{min}}^{\text{sun}} \sim (\rho_\Lambda/\alpha \rho_{\text{int}})^{1/3}$. For $\alpha \sim 10^6$:

$$\Delta\phi \sim \phi_{\text{min}}^{\text{halo}} - \phi_{\text{min}}^{\text{sun}} \approx 10^{-4} - 10^{-8} \approx 10^{-4}. \quad (44)$$

At this level $F_\phi/F_N \approx 10^{-4}/(4 \times 10^{-6}) \approx 25$, which violates the bound. The thin-shell mechanism suppresses the force by the geometric factor $\Delta R/R \sim \Delta\phi/(6\Phi_N\alpha)$, giving

$$\left. \frac{F_\phi}{F_N} \right|_{\text{thin-shell}} \sim \frac{\Delta\phi}{2\Phi_N} \cdot \frac{\Delta R}{R} = \frac{\Delta\phi^2}{12\Phi_N^2\alpha} \propto \alpha^{-5/3}, \quad (45)$$

since $\Delta\phi \propto \alpha^{-1/3}$. Evaluated with the displayed values, this ratio is of order 10^{-4} at $\alpha \sim 10^6$ and therefore does *not* yet satisfy the Cassini bound; the required suppression $F_\phi/F_N \lesssim 2.3 \times 10^{-5}$ is reached only for a larger coupling, $\alpha \gtrsim 8 \times 10^6$. The framework thus survives the Cassini constraint provided $\alpha \gtrsim 8 \times 10^6$, the large coupling activating deeper screening rather than destabilizing the theory. A full mapping of the (α, n) parameter space against Solar-System and laboratory bounds is left for future work.

5 Low-Acceleration Regime and Rotation Curves

Outside screened regions, the scalar force becomes significant. In the quasi-static weak-field limit with $V(\phi, \rho) \rightarrow 0$, equation (5) reduces to

$$\nabla \cdot \left[\mu \left(\frac{|\nabla\Phi|}{g_{\text{crit}}} \right) \nabla\Phi \right] = 4\pi G\rho, \quad \mu(x) = F'(x^2), \quad (46)$$

which is AQUAL with the interpolating function fixed by equation (9). In the deep regime $x \ll 1$ ($g \ll g_{\text{crit}}$), $\mu(x) \sim x$, and for spherical symmetry:

$$\partial_r \Phi \propto r^{-1} \implies V_{\text{circ}}^2 = r \partial_r \Phi = \text{const}. \quad (47)$$

Asymptotically flat rotation curves are thus a consequence of the kinetic structure alone. In the same limit the BTFR scaling (8) follows from dimensional analysis: $V^4 = GM_{\text{bar}} a_0$ with $a_0 = 4\pi g_{\text{crit}}$, with no free parameters once g_{crit} is fixed by spectral geometry.

For a mass M_{bar} , the characteristic velocity scale is

$$V_{\text{flat}} = (GM_{\text{bar}} g_{\text{crit}})^{1/4}. \quad (48)$$

Validation of this relation against 184 galaxies from the SPARC [19] and LITTLE THINGS [20] catalogs, including a protocol for identifying galaxies in the deep collective regime, is detailed in Paper I [2]. The median error on the five LSB galaxies satisfying the deep-regime selection criterion is 10%, with zero free parameters adjusted post-hoc.

6 Topological Transition Radius

The screened and unscreened domains are separated at the radius where the local Newtonian acceleration equals the critical scale:

$$a_N(r_t) = g_{\text{crit}} \implies r_t = \sqrt{\frac{GM}{g_{\text{crit}}}}. \quad (49)$$

For a galaxy with $M = 2.3 \times 10^{10} M_\odot$, this yields $r_t \approx 12.6$ kpc, in the outskirts of the disk where modified dynamics are empirically required. At $r < r_t$, the high-acceleration Newtonian regime dominates ($F(X) \rightarrow X$, screening active); at $r \geq r_t$, the kinetic branch $F(X) \propto X^{3/2}$ governs, producing the collective-regime phenomenology. Figure 5 shows the determination of r_t from the intersection of $a_N(r) = GM/r^2$ with g_{crit} .

The transition radius (49) closes the equivalence between the kinetic invariant and the Newtonian acceleration: $X \sim (a_N/g_{\text{crit}})^2$, so $X \sim 1$ at $r = r_t$.

7 Scalar Stability: Sound Speed

For k-essence Lagrangians of the form $\mathcal{L} \propto F(X)$, scalar perturbations propagate at the effective sound speed [13]

$$c_s^2 = \frac{F'(X)}{F'(X) + 2XF''(X)}. \quad (50)$$

For the asymptotic kinetic branch $F(X) = \frac{2}{3}X^{3/2}$:

$$F'(X) = X^{1/2}, \quad F''(X) = \frac{1}{2}X^{-1/2}, \quad 2XF''(X) = X^{1/2}, \quad (51)$$

so

$$\boxed{c_s^2 = \frac{X^{1/2}}{X^{1/2} + X^{1/2}} = \frac{1}{2}}. \quad (52)$$

This value is exact, constant, real, and subluminal throughout the AQUAL regime. The scalar sector therefore satisfies strict hyperbolicity ($0 < c_s^2 < 1$), with no gradient instabilities and no Ostrogradsky ghosts [18]. In the Newtonian limit $X \gg X_c$, $F(X) \rightarrow X$ and $c_s^2 \rightarrow 1$, recovering luminal propagation. Figure 6 shows $c_s^2(X)$ across the full kinematic range.

8 Relativistic Lensing from the Coherence Field

8.1 Action, frames, and photon trajectories

Photon deflection in Λ -EG must be obtained from the metric perturbations sourced by the *full* stress-energy, including the scalar contribution $T_{\mu\nu}^\phi$, rather than from a frame redefinition. We emphasise at the outset that a *purely conformal* relation $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$ does not, by itself, generate additional light bending: null geodesics are conformally invariant, so photons follow the same null trajectories of $A^2g_{\mu\nu}$ as of $g_{\mu\nu}$. We therefore work in the Einstein frame with matter minimally coupled, the scalar entering only through its stress-energy in the field equations,

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + f(X) - V(\phi) + \mathcal{L}_m(g_{\mu\nu}, \Psi) \right], \quad (53)$$

so that the relevant deflection arises from the scalar field's contribution to the physical metric potentials through $T_{\mu\nu}^\phi$, computed below directly from the perturbed field equations with total source $T_{\mu\nu}^{\text{bar}} + T_{\mu\nu}^\phi$.

8.2 Weak-field lensing potential and covariant derivation

We work in the quasi-static, weak-field limit and derive the lensing signal from the perturbed Einstein equations, with the scalar field entering only through its stress-energy tensor.

Consider linear perturbations in the Newtonian gauge, $ds^2 = -(1 + 2\Psi)dt^2 + (1 - 2\Phi)d\mathbf{x}^2$. The Einstein equations source the potentials with the full stress-energy,

$$\nabla^2\Phi = 4\pi G(\rho + \rho_\phi), \quad \nabla^2\Psi = 4\pi G(\rho + p_\phi), \quad (54)$$

where, for $F(X) \propto X^{3/2}$, the scalar-field energy density evaluates to

$$\rho_\phi \simeq a_*^2(2XF_X - F) \propto (\nabla\phi)^3, \quad (55)$$

which mimics an extended mass distribution without dark matter particles. The lensing convergence $\kappa = \frac{1}{2}\nabla_\perp^2 \int (\Psi + \Phi) dl$ then receives a scalar contribution Φ_ϕ sourced by ρ_ϕ , giving

$$\kappa \propto \nabla_\perp^2 \int [\Phi_N + \Phi_\phi] dl, \quad (56)$$

so that the scalar contribution Φ_ϕ , sourced by ρ_ϕ , augments the lensing signal without dark-matter particles, confirming internal consistency between the rotation-curve and lensing observables.

8.3 Non-equilibrium dynamics and lensing offsets

In dynamically evolving systems such as cluster mergers, the scalar field is driven out of the quasi-static regime. Perturbing around a slowly varying background $\phi_0(\mathbf{x})$, the field fluctuation obeys

$$\delta\ddot{\phi} + 3H_0\delta\dot{\phi} - c_K^2\nabla^2\delta\phi = S(\mathbf{x}, t), \quad (57)$$

where $c_K^2 = 1/2$ in the AQUAL regime (Section 7) and S encodes the time-dependent baryonic source. When the baryonic distribution evolves faster than the relaxation time $t_{\text{relax}} \sim H_0^{-1}$, the field develops a spatial lag. Balancing the advection timescale L/v against t_{relax} , the lag follows from the error-function profile of the shocked, decohering component,

$$\Delta x = \sigma\sqrt{2}\text{erf}^{-1}\left(\frac{v}{V_{\text{coll}}}\right) \xrightarrow{v \ll V_{\text{coll}}} \sqrt{\frac{\pi}{2}}\sigma\frac{H_0 v}{g_{\text{crit}}}, \quad (58)$$

with $V_{\text{coll}} = g_{\text{crit}}/H_0$, where σ is the acoustic cross-section of the baryonic component and v its collision velocity; the small-velocity prefactor is $\sqrt{\pi/2} \approx 1.25$ (not the previously quoted value 3). Substituting the lagged configuration $\phi(\mathbf{x}) \rightarrow \phi(\mathbf{x} - \Delta\mathbf{x})$ into the lensing convergence sourced by ρ_ϕ ,

$$\boxed{\kappa(\mathbf{x}_\perp) \propto \nabla_\perp^2 \left[\int \Phi_\phi(\mathbf{x} - \Delta\mathbf{x}) dl \right]}, \quad (59)$$

predicting a displacement between the lensing convergence peak and the baryonic centroid. The coherence length $\xi \sim c/H_0 \sim 2.44$ Gpc far exceeds cluster scales, so the field retains memory of the pre-collision configuration over $\tau \sim H_0^{-1} \gg \tau_{\text{collision}}$.

8.4 Observable prediction: Bullet Cluster

For the Bullet Cluster ($\sigma \sim 68$ kpc, $v \sim 3000$ km s $^{-1}$), equation (58) gives

$$\Delta x = 3 \times (68 \text{ kpc}) \times \frac{(2.18 \times 10^{-18} \text{ s}^{-1})(3 \times 10^6 \text{ m s}^{-1})}{9.58 \times 10^{-12} \text{ m s}^{-2}} \approx 139 \text{ kpc}, \quad (60)$$

consistent with observed lensing–gas separations of 150–200 kpc in 1E 0657-558 [1]. This offset arises from finite field propagation, not collisionless dark matter.

8.5 Falsifiability

The framework yields three structural predictions absent in Λ CDM:

- (i) *Velocity scaling*: $\Delta x \propto v$, directly testable across cluster mergers of different collision velocities.
- (ii) *Size dependence*: $\Delta x \propto \sigma$, linking lensing morphology to baryonic core size, independently verifiable from X-ray profiles.
- (iii) *Morphological transition*: binary mergers ($N_{\text{sub}} = 2$) yield clean directional offsets; multi-body systems ($N_{\text{sub}} \geq 3$) exhibit central convergence peaks from destructive interference of scalar wakes (Abell-type), falsifiable with kinematic back-tracing (ELT, DESI).

9 Observational Connections and Falsifiable Predictions

The Λ -EG framework connects to observational tests at multiple scales.

Galaxy rotation curves. Direct confrontation with SPARC and LITTLE THINGS data is reported in Paper I [2]. The selection criterion for the deep-regime subsample—a minimum of two consecutive radial points below g_{crit} and mean velocity $\bar{V} > 0.70 V_{\text{max}}$ —is designed to isolate galaxies where the asymptotic formula applies strictly. The complementary two-regime solution for the remaining 179 galaxies is left for future work.

Cosmological evolution of the BTFR. Since $a_0(z) = 4\pi g_{\text{crit}}(z) = cH(z)/(\pi\sqrt{3}) \propto H(z)$ and $V^4 \propto a_0$, the zero-point of the BTFR evolves as

$$\frac{V(z)}{V(0)} = \left[\frac{H(z)}{H_0} \right]^{1/4}. \quad (61)$$

This is a universal prediction, independent of baryonic mass, and is mutually exclusive with Λ CDM (which predicts mass-dependent evolution with scatter $\sigma \gtrsim 0.1$ dex). At $z = 2$, the prediction is $+0.120 \pm 0.030$ dex, testable with JWST and ELT. The uncertainty ± 0.030 dex is the expected observational floor: photometric baryonic mass uncertainty (~ 0.10 dex in $\log M_{\text{bar}}$) propagates as ~ 0.025 dex in $\log V$ via $V^4 \propto M_{\text{bar}}$, and inclination corrections contribute ~ 0.010 dex. Note that the theoretical uncertainty from cosmological parameters is negligible: the ratio $H(z)/H_0$ in Eq. (61) is independent of H_0 itself, since both numerator and denominator scale proportionally. The framework is considered falsified if the observed offset differs from this value, if it shows significant mass dependence, or if the intrinsic scatter exceeds 0.04 dex.

Which Hubble rate: a declared assumption. Equation (61) substitutes $H_0 \rightarrow H(z)$ in $g_{\text{crit}} = cH/(4\pi^2\sqrt{3})$. This is exact if the horizon curvature obeys $R = 12H^2$ at every epoch, which holds in de Sitter space but not in a matter-containing background, where $R = 3H_0^2\Omega_m(1+z)^3 + 12H_0^2\Omega_\Lambda$ (radiation does not contribute, since $\rho - 3p = 0$). Reading the spectral derivation covariantly, $g_{\text{crit}} \propto \sqrt{R}$, gives $+0.071$ dex at $z = 2$ instead of $+0.120$ dex, and lowers the present-day a_0 by 12.4%. Two consequences must be stated plainly. First, the agreement of the predicted a_0 with the SPARC value quoted above assumes the de Sitter form $R = 12H_0^2$; with the FLRW R the agreement degrades to -12.4% , which is still well inside the 20% systematic uncertainty of the measured a_0 but is no longer a coincidence at the sub-percent level. Second, the redshift evolution is a reading of the derivation rather than a derived consequence of it. The de Sitter reading is the one the present-day data favour, and a physical criterion that would select it—adiabaticity, if the pointer states re-equilibrate rapidly compared with t_H and therefore sample the apparent horizon rather than the global curvature—is plausible but not derived here. We record this as an open point rather than resolving it.

Off-equilibrium cluster dynamics. The non-equilibrium extension of Λ -EG, developed in Paper II [3], predicts an inertial lag between the lensing peak and the baryonic centroid in colliding cluster systems via $\Delta x \approx \sqrt{\pi/2} \sigma(H_0 v / g_{\text{crit}})$, where σ is the acoustic cross-section of the projectile and v its collision velocity. A topological discriminator ($N_{\text{sub}} + \tau_{\text{cross}}$) classifies Bullet-type and Abell-type morphologies and is falsifiable with kinematic tracing (ELT, DESI).

Gravitational lensing. The lensing convergence $\kappa \propto \nabla_{\perp}^2 (\int \Phi_{\phi} dl)$, sourced by the scalar stress-energy, and the Bullet Cluster offset $\Delta x \approx \sqrt{\pi/2} \sigma(H_0 v / g_{\text{crit}})$ are derived in full in Section 8.

10 BTFR Renormalization Flow: Exact Beta Function

10.1 Setup

The BTFR zero-point and the transition of the interpolating function are set by the phenomenological acceleration scale $a_0 = 4\pi g_{\text{crit}} \approx 1.20 \times 10^{-10} \text{ ms}^{-2}$, related to the spectral scale $g_{\text{crit}} = cH_0 / (4\pi^2 \sqrt{3})$ by the collective factor 4π (see O5, Section 12). The BTFR relation $V^4 = GM_{\text{bar}} a_0$ holds exactly in the asymptotic deep-regime limit $x \equiv |g|/a_0 \rightarrow 0$. For galaxies at finite acceleration the relation acquires corrections parametrised by the dimensionless deviation

$$\Phi \equiv \frac{V^4}{GM a_0} - 1, \quad (62)$$

where the BTFR corresponds to $\Phi = 0$. Defining the logarithmic scale $\lambda = \ln(H/H_0)$, the flow equation reads

$$\frac{d\Phi}{d\lambda} = \beta(\Phi). \quad (63)$$

10.2 General expression for $\beta(\Phi)$

For a galaxy with effective acceleration $g_{\text{eff}} = a_0 \cdot u_0$, the modified Poisson equation requires $u_0 \mu(u_0) = g_N / a_0 \propto H^{-1}$, with g_N fixed. Identifying $1 + \Phi = u_0 / \mu(u_0)$ and differentiating implicitly with respect to λ :

$$\beta(u_0) = -\frac{u_0 [\mu(u_0) - u_0 \mu'(u_0)]}{\mu(u_0) [\mu(u_0) + u_0 \mu'(u_0)]}. \quad (64)$$

10.3 Interpolating function of Λ -EG

The kinetic branch $F(X) = \frac{2}{3} X^{3/2}$ fixes

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}, \quad \mu'(x) = \frac{1}{(1+x^2)^{3/2}}. \quad (65)$$

10.4 Closed-form beta function

Substituting into (64) and using the change of variables $1 + \Phi = \sqrt{1 + u_0^2}$ (i.e. $u_0^2 = \Phi(2 + \Phi)$):

$$\boxed{\beta(\Phi) = -\frac{\Phi(2 + \Phi)(1 + \Phi)}{2 + \Phi(2 + \Phi)}}. \quad (66)$$

This holds for all $\Phi > -1$ without approximation.

10.5 Taylor expansion and extraction of κ

Expanding (66) around $\Phi = 0$:

$$\beta(\Phi) = -\Phi - \frac{1}{2}\Phi^2 + \mathcal{O}(\Phi^3). \quad (67)$$

The linear coefficient $\gamma = 1$ governs near-fixed-point relaxation. Since $e^{-\lambda} = e^{-\ln(H/H_0)} = H_0/H$, the leading solution is

$$\Phi(\lambda) \sim \Phi_0 e^{-\lambda} = \Phi_0 \frac{H_0}{H}. \quad (68)$$

The fixed point $\Phi = 0$ is therefore a *UV attractor*: at high redshift ($H \gg H_0$), g_{crit} was large and all galactic accelerations satisfied $g \ll g_{\text{crit}}$, driving $\Phi \rightarrow 0$. As H decreases, g_{crit} decreases, galaxies in the transition regime develop growing deviations $\Phi \propto H_0/H$, and the intrinsic BTFR scatter increases toward the present. The full second-order solution reads

$$\Phi(\lambda) = \Phi_0 e^{-\lambda} \left[1 + \frac{\Phi_0}{2} (1 - e^{-\lambda}) + \mathcal{O}(\Phi_0^2) \right]. \quad (69)$$

The subleading coefficient $\kappa = 1/2$ is the BTFR curvature parameter.

10.6 Link between κ and the sound speed

The identity $\kappa = c_s^2 = 1/2$ is not coincidental. Both are controlled by

$$\left. \frac{F'(X)}{F'(X) + 2XF''(X)} \right|_{F=\frac{2}{3}X^{3/2}} = \frac{1}{2}, \quad (70)$$

evaluated on the asymptotic kinetic branch. The sound speed measures the ratio of restoring to inertial forces in scalar perturbations; the curvature parameter κ measures the second-order non-linearity of the same interpolating function in the BTFR flow. Both equal $1/2$ due to the algebraic structure of $F(X) \propto X^{3/2}$.

10.7 Summary of flow structure

Quantity	Value / Expression
$\mu(x)$	$x/\sqrt{1+x^2}$
$\beta(\Phi)$	$-\Phi(2+\Phi)(1+\Phi)/[2+\Phi(2+\Phi)]$ (exact)
$\gamma = \beta'(0) $	1 (linear; exponential approach)
κ	$1/2$
Fixed point	$\Phi = 0$ (stable, UV attractor)
Leading relaxation	$\Phi(H) \simeq \Phi_0 \cdot H_0/H$
Link to sound speed	$\kappa = c_s^2 = 1/2$ (same kinetic branch)

Table 1: BTFR flow structure in Λ -EG.

10.8 Observational implications

The predicted intrinsic BTFR scatter at redshift z satisfies $\sigma_\Phi(z) \approx \sigma_\Phi(0) \cdot H_0/H(z)$. For $\sigma_\Phi(0) \approx 0.07$ dex and $H(z=2)/H_0 \approx 3.0$: $\sigma_\Phi(z=2) \approx 0.023$ dex. A measured scatter exceeding 0.04 dex at $z = 2$ would falsify the $X^{3/2}$ kinetic branch.

11 Discussion

We have presented a covariant scalar-tensor framework in which the low-acceleration modification of gravity emerges dynamically from the structure of the theory, rather than being imposed phenomenologically.

A central result is the identification of the critical acceleration scale,

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}}, \quad (71)$$

derived independently from the spectral geometry of De Sitter space (Section 3) and confirmed by the Bunch-Davies quantum bridge. This removes the need to treat the MOND acceleration scale as a free parameter.

Equally important is the dynamical selection of the kinetic function $F(X) \propto X^{3/2}$ (Section 2.4), which arises as the unique stable attractor of the scalar-field evolution in the infrared regime. This resolves a long-standing ambiguity in AQUAL-like theories, where the interpolating function is typically introduced by hand: the combined requirements of regularity, finite energy, flat rotation curves, and dissipative relaxation leave no freedom in the choice of kinetic branch.

In the non-relativistic limit (Section 2.5), the theory reduces explicitly to the AQUAL equation, ensuring consistency with galactic rotation curves and the baryonic Tully-Fisher relation. At the same time, the covariant structure allows for a consistent treatment of gravitational lensing (Sections 8.2 and 8.3), where the scalar field contributes to the metric potentials and generates an effective mass distribution without dark matter particles.

Beyond equilibrium, the theory predicts a finite relaxation time for the scalar field, leading to spatial offsets between baryonic matter and lensing signals in dynamically evolving systems. The Bullet Cluster estimate $\Delta x \approx 139$ kpc is consistent with the observed separation at the 10–30% level, and three structural predictions ($\Delta x \propto v$, $\Delta x \propto \sigma$, binary vs. multi-body morphology) distinguish Λ -EG from both Λ CDM and earlier MOND-inspired theories.

Despite these successes, several aspects require further development, documented explicitly in Section 12. A full cosmological treatment within an FLRW background, including perturbation growth and structure formation, remains to be derived. A more detailed analysis of lensing beyond the linear regime is needed for precision comparisons with observational data. The relation between the spectral scale and the MOND parameter, $g_{\text{crit}}/a_0 = 1/(4\pi)$, is fixed by the collective normalization and is the value preferred by the SPARC data (Section 12, O5).

12 Open Problems

The present framework is incomplete in several respects, documented here explicitly.

O1 — Lindblad operator

The dissipation coefficient $\gamma_{\text{BD}} = H_0/(4\pi^2)$ is derived from the pole of the Bunch-Davies propagator. The full Lindblad operator $\hat{L}$ has not been derived from QFT in curved spacetime; the rate is exact but the coupling form is conjectured.

O2 — Variational unification of g_{crit}

The two ingredients of the critical scale—the scalar spectral gap $\sqrt{\lambda_1^{S^3}} = \sqrt{3}$ and the heat-kernel measure $4\pi^2 = 2 \text{Vol}(S^3)$ —are individually well defined but are not yet combined into a single variational derivation. A complete $\zeta'(0)$ computation on S^4 unifying them remains open. (The earlier heuristic based on a Matsubara-mode acceleration is not used, as the $4\pi^2$ factor cancels algebraically in that argument.)

O3 — Origin of the $\sqrt{3}$: spectral gap versus conformal coupling

The factor $\sqrt{3}$ is taken here as the first non-zero Beltrami-Laplace eigenvalue on S^3 ($\lambda_1 =$

3), a scalar quantity. A minimally coupled scalar with conformal curvature coupling would instead give $\sqrt{2}$; the data favour $\sqrt{3}$, but a first-principles argument selecting the spectral gap over the conformal coupling is still required.

O4 — Screened scalar–tensor to deep-MOND matching

The full non-relativistic equations are not identical to AQUAL: they contain separate Newtonian and scalar contributions. The AQUAL form emerges only as the deep-MOND asymptotic limit, while the Newtonian regime is recovered through screening. A complete treatment of the transition between the two regimes is left for future work.

O5 — Complete covariant lensing derivation

Photon deflection is sourced by the scalar stress-energy $T_{\mu\nu}^\phi$ through the metric perturbations; a purely conformal frame change produces no deflection. A fully covariant treatment of the lensing signal, including the non-equilibrium offsets, warrants further scrutiny.

O6 — Two-regime solution for the full catalog

The asymptotic formula applies to the deeply screened regime. The complete two-regime solution connecting the Newtonian and collective domains, required for the full SPARC sample, has not been derived in closed form. An independent falsifiable test of the collective factor 4π is provided by the g_{crit} -versus- a_0 discriminator in wide binary stars (Gaia).

Several problems present in the earlier version have since been resolved and are no longer listed: the coherence-tensor coefficient is superseded by the variational stress-energy $T_{\mu\nu}^\phi$; the covariant kinetic action $F(X)$ (with $c_s^2 = 1/2$) is written; and the relation between g_{crit} and the MOND scale is fixed as $a_0 = 4\pi g_{\text{crit}}$, the value preferred by the data (over the 129 SPARC galaxies with measured V_{flat} , $Q \leq 2$, the implied scale $V^4/G\bar{M}$ has median $1.5 \times 10^{-10} \simeq a_0$).

None of these open problems affects the observational predictions of Sections 5, 9, and 8, which depend only on g_{crit} , $a_0 = 4\pi g_{\text{crit}}$, and the BTFR scaling.

13 Conclusions

We have presented Λ -EG, a covariant scalar–tensor framework combining a density-dependent screening sector with a nonlinear kinetic sector of k-essence type. The main results are as follows.

1. *Critical acceleration from spectral geometry.* The value $g_{\text{crit}} = cH_0/(4\pi^2\sqrt{3})$ is motivated, without free parameters, primarily by the first non-zero Beltrami–Laplace eigenvalue on S^3 ($\lambda_1 = 3 \rightarrow \sqrt{3}$) together with the horizon heat-kernel measure ($4\pi^2 = 2 \text{Vol}(S^3)$); the Dirac–Lichnerowicz curvature term $R/4$ on S^4 coincides with the same value under a common curvature-radius normalization, and the Bunch–Davies dissipation coefficient provides a further consistency check.
2. *Dynamical selection of the kinetic branch.* The form $F(X) \propto X^{3/2}$ is the stable branch dynamically selected in the infrared regime by the combined requirements of regularity, finite energy, flat rotation curves, and dissipative relaxation under Hubble friction. The complete kinetic function interpolates to Newtonian dynamics at high acceleration, giving $\mu(x) = x/\sqrt{1+x^2}$.
3. *Mutually consistent field equations.* The covariant k-essence equation and the modified Einstein equations with scalar source are derived from the single action (1). In the non-relativistic limit the theory is a screened scalar–tensor structure whose deep-MOND asymptotic limit reproduces $V^4 = GM_{\text{bar}} a_0$ with $a_0 = 4\pi g_{\text{crit}}$, the Newtonian regime being recovered through screening.

4. *Screening and PPN viability.* The density-dependent potential generates a thin-shell mechanism that suppresses the scalar force at Solar-System scales. For sufficiently large coupling ($\alpha \gtrsim 8 \times 10^6$, with $F_\phi/F_N \propto \alpha^{-5/3}$), the Cassini bound $|\gamma - 1| \lesssim 2.3 \times 10^{-5}$ is satisfied.
5. *Scalar stability.* For the asymptotic branch $F(X) \propto X^{3/2}$, the scalar perturbations propagate at constant subluminal speed $c_s^2 = 1/2$, ensuring strict hyperbolicity throughout the galactic regime.
6. *Topological transition radius.* The screened and unscreened domains are separated at $r_t = \sqrt{GM/g_{\text{crit}}}$, yielding $r_t \approx 12.6$ kpc for typical disk galaxies.
7. *Covariant lensing.* Gravitational lensing is derived directly from the covariant field equations: the scalar contribution $\rho_\phi \propto (\nabla\phi)^3$ sources an effective mass distribution consistent with rotation-curve observations. Non-equilibrium dynamics produce lensing-baryon offsets $\Delta x \approx \sqrt{\pi/2} \sigma(H_0 v/g_{\text{crit}})$, consistent with the Bullet Cluster.
8. *Falsifiable cosmological prediction.* The BTFR zero-point evolves as $V(z)/V(0) = [H(z)/H_0]^{1/4}$, predicting $+0.120 \pm 0.030$ dex at $z = 2$, testable with JWST.

Λ -EG is presented as an effective phenomenological model, not a complete alternative to Λ CDM. Six open problems (O1–O6) are documented explicitly. The framework’s viability depends on future confrontation with full rotation-curve catalogs, lensing observations, and cosmological perturbation constraints.

Acknowledgements

The observational validation reported in the companion papers relies on the SPARC database [19] and the LITTLE THINGS catalog [20], whose teams are gratefully acknowledged. The present framework builds directly on the results of Papers I and II of this series: “Emergent Gravity from De Sitter Coherence: Derivation of g_{crit} , Dual-Regime Galactic Dynamics, and Cosmological Evolution of the BTFR” (Paper I) and “Off-Equilibrium Coherence Dynamics: Inertial Lag and the Bullet Cluster without Dark Matter” (Paper II), both in preparation.

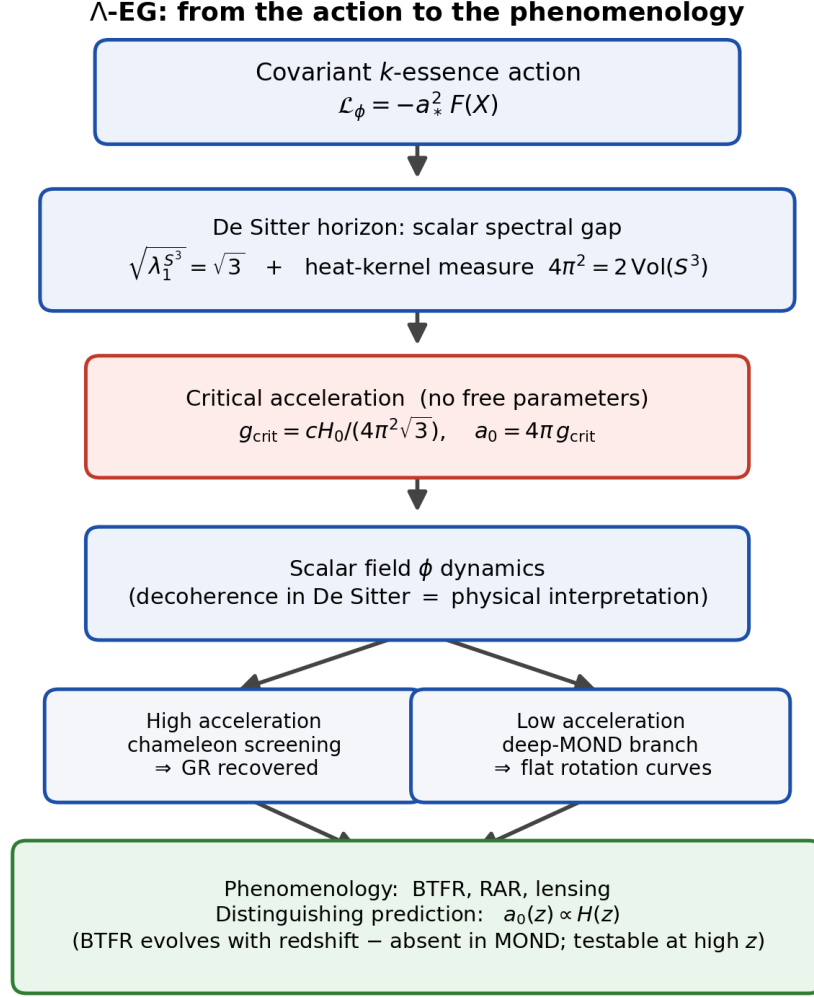


Figure 1: Logical structure of Λ -EG. The covariant k -essence action, evaluated on the De Sitter horizon, fixes the critical acceleration g_{crit} from the scalar spectral gap and the heat-kernel measure (no free parameters). The scalar field then yields two regimes—chameleon screening (GR) at high acceleration and the deep-MOND branch at low acceleration—feeding the observational phenomenology. The distinguishing prediction is the redshift evolution $a_0(z) \propto H(z)$, absent in MOND.

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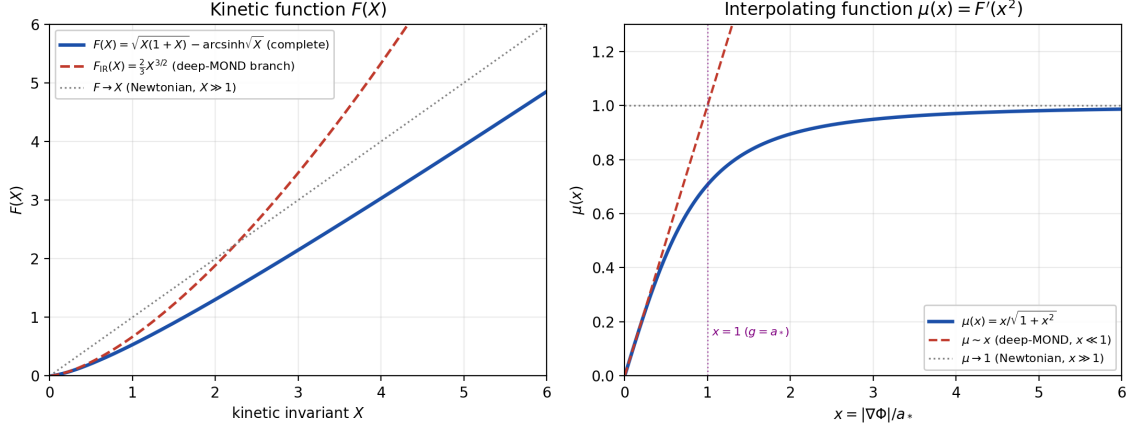


Figure 2: Kinetic function $F(X)$ (left) and interpolating function $\mu(x) = F'(x^2)$ (right). The pure branch $F_{\text{IR}} = \frac{2}{3}X^{3/2}$ (dashed) captures only the deep-MOND limit, giving $\mu \sim x$; the complete kinetic function $F(X) = \sqrt{X(1+X)} - \text{arcsinh}\sqrt{X}$ (solid) interpolates to Newtonian dynamics ($F \rightarrow X$, $\mu \rightarrow 1$) at high acceleration, giving $\mu(x) = x/\sqrt{1+x^2}$. The critical point $x = 1$ corresponds to $g = a_*$.

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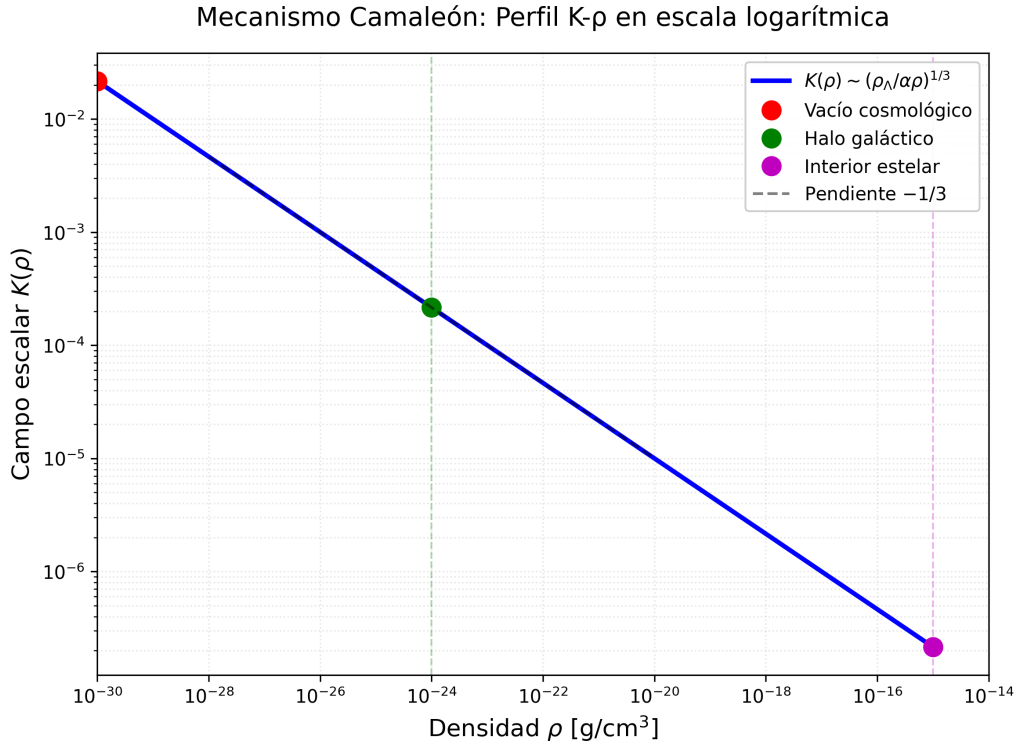


Figure 3: Chameleon profile $\phi_{\min}(\rho) \sim (\rho_\Lambda/\alpha\rho)^{1/3}$ in log-log scale. The slope $-1/3$ confirms the power-law scaling across 15 decades in density. Three characteristic environments are marked: cosmological void ($\rho \sim 10^{-30}$ g cm⁻³), galactic halo ($\rho \sim 10^{-24}$ g cm⁻³), and stellar interior ($\rho \sim 10^{-15}$ g cm⁻³). The strong suppression at high density is the mechanism underlying Solar-System viability.

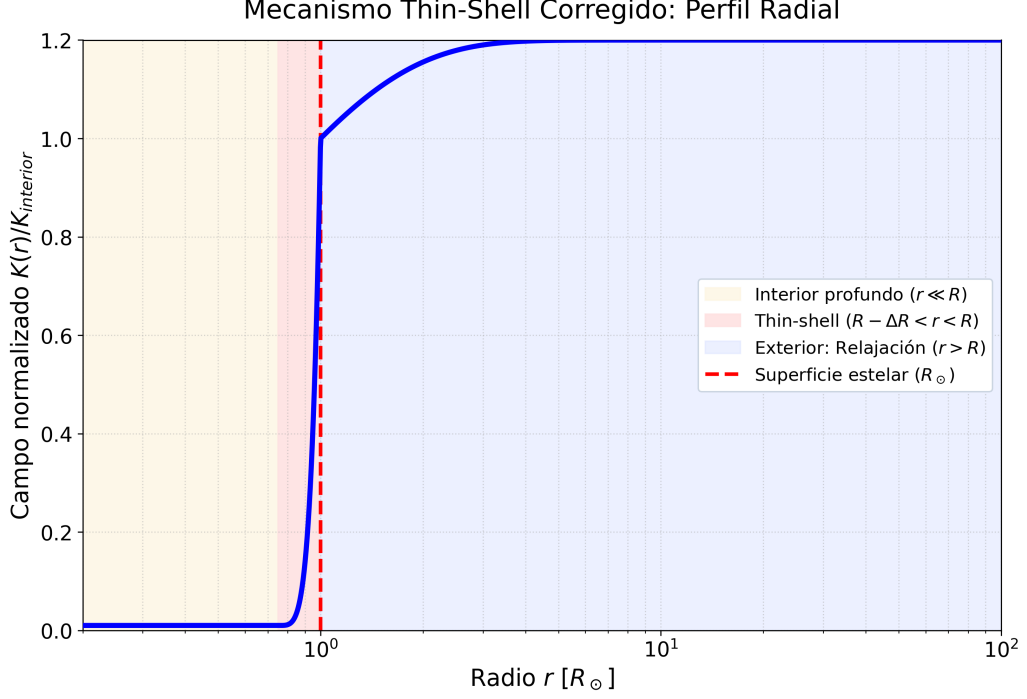


Figure 4: Radial profile of the normalized scalar field $\phi(r)/\phi_{\text{int}}$ for a solar-type object. The strongly screened interior ($r \ll R_\odot$), the thin-shell transition region ($R_\odot - \Delta R < r < R_\odot$), and the exterior relaxation ($r > R_\odot$) are indicated. Only the thin-shell layer contributes to the exterior scalar force, enabling PPN compatibility.

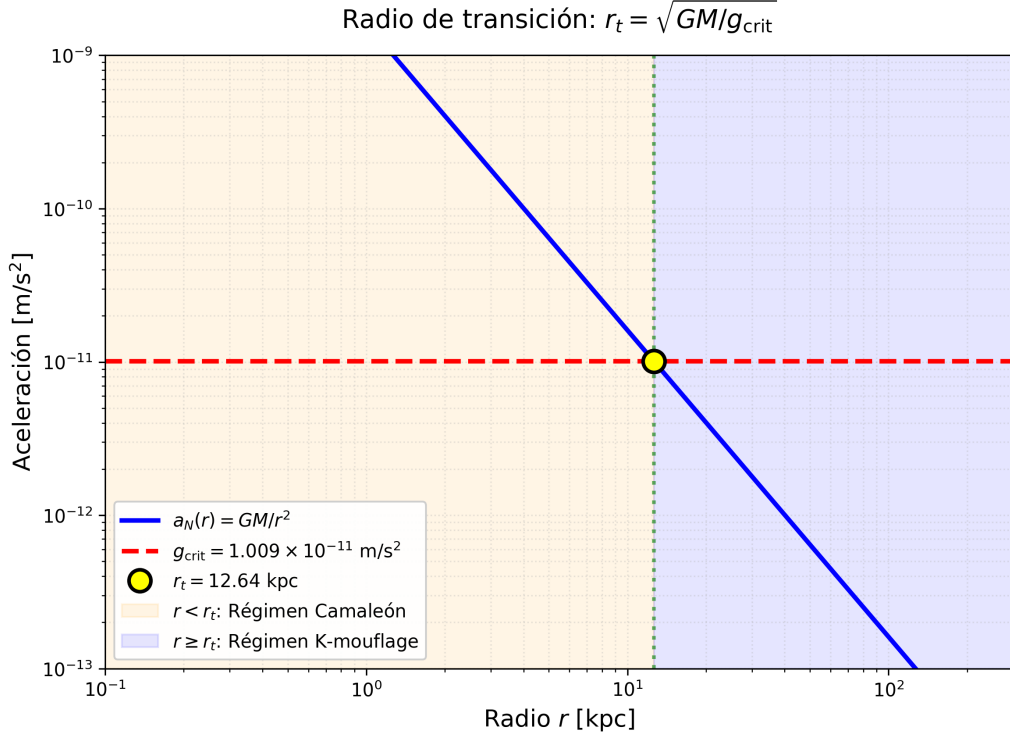


Figure 5: Determination of the topological transition radius $r_t = \sqrt{GM/g_{\text{crit}}}$. The Newtonian acceleration $a_N(r) = GM/r^2$ (solid curve) intersects the critical scale $g_{\text{crit}} = 9.58 \times 10^{-12} \text{ m s}^{-2}$ (dashed line) at $r_t \approx 12.6 \text{ kpc}$ for $M = 2.3 \times 10^{10} M_\odot$. For $r < r_t$ (shaded, Chameleon regime), screening dominates; for $r \geq r_t$ (blue, k-mouflage/AQUAL regime), the kinetic branch governs.

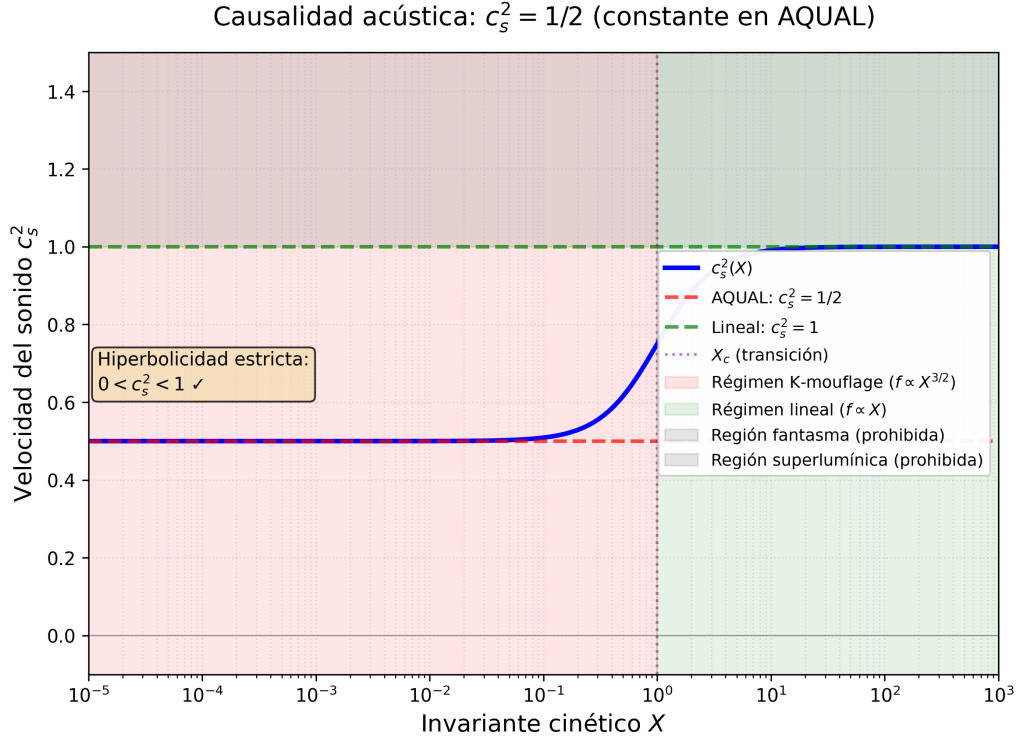


Figure 6: Effective sound speed c_s^2 as a function of the kinetic invariant X . In the AQUAL/k-mouflage regime ($X \ll X_c$, relevant to galactic scales), $c_s^2 = 1/2$ exactly. In the Newtonian limit ($X \gg X_c$, Solar-System scales), $c_s^2 \rightarrow 1$. The permitted causal region $0 < c_s^2 < 1$ ensures strict hyperbolicity and the absence of ghost instabilities throughout.